

The Open Data Agent Platform

A Standard for Governed AI-to-Data Access

White Paper • Version 1.0 • July 2026

Maintained by the Open Data Agent Platform community

github.com/open-data-agent-platform • Initiating signatory: GoogleSQL.com

License: CC0 (standard texts) • Apache-2.0 (reference implementation)

Abstract

AI agents now write more SQL than the humans who once supervised every query. Generation has been solved — spectacularly — but the layer that made enterprise data access trustworthy for fifty years has been deleted along the way: the accountable human in the loop. This paper defines the Open Data Agent Platform (ODAP), a vendor-neutral standard for the gateway through which AI agents must access enterprise data. It establishes five normative principles (the Five Musts), a five-layer reference architecture, a public clause-level scoring rubric, and the governance process by which the standard itself evolves. It reports the first public application of the rubric to five major vendors (the ODAP Landscape 2026), documents the standard's first institutional adoption, and describes an open-source reference implementation with an honest clause-by-clause conformance mapping. The central claim is simple: in the agentic era, the scarce resource is not intelligence — it is trust, and trust is built at the gateway.

1. The Moment We Are In

For fifty years, every query that touched enterprise data passed through a human being — a person who understood the schema, feared the bill, and answered for the result. That person was slow and imperfect, but accountable. Accountability was the invisible infrastructure on which everything else stood.

That era has ended. AI agents write SQL faster than humans can read it, and within a few years machines will produce more queries than every human in history combined. The consequences are already visible in production environments: bills that arrive as surprises because no one estimated the query before it ran; permission violations that no policy engine saw because the policy lived outside the generation path; plausible-but-wrong answers delivered confidently to decision-makers; audit trails that terminate at the unhelpful sentence "the AI did it."

Industry has begun to respond. At the 2026 Data + AI Summit, the chief executive of one major platform vendor described governance as "the gate that decides whether you get to scale at all," and shipped hard spend caps

and pre-execution approvals as flagship capabilities. Research from MIT and Microsoft (the Murakkab system) demonstrated that agentic workflows can meet identical requirements using roughly 35% of the computation, 27% of the energy, and under 25% of the cost of conventional deployments — establishing that resource governance of agent workflows is not only necessary but tractable. Regulators have moved as well: under the EU AI Act, data governance functions are shifting from optional to mandated, and governance-related roles are among the fastest-growing data specializations.

Each of these signals points at the same missing layer. Vendors are shipping pieces of it; regulators are demanding outcomes from it; enterprises are hiring for it without a shared definition of what "it" is. What has been absent is a vendor-neutral standard that names the layer, defines its obligations, and makes conformance measurable. That is the purpose of this document.

2. The Five Musts

The ODAP doctrine can be stated in one sentence:

When an AI agent touches enterprise data, it must pass through a single gateway that is open, semantic, governed, auditable, and cost-controlled.

These are the Five Musts. They are normative — a platform that satisfies four of the five is not an ODAP.

Must 1 — Open. The gateway must speak open table formats (Apache Iceberg, Delta Lake, Apache Hudi), open protocols, and heterogeneous sources: warehouses, lakes, object storage, and operational databases. Enterprise data will never again live in one place. A gateway that locks its users in is not a gateway; it is a wall with a toll booth. Openness includes exit: semantic definitions, policies, and audit history must be exportable in documented formats.

Must 2 — Semantic. Agents must query meaning, not tables. A business metric such as "monthly active users" must be defined once, in business language, by the people who own it — and must resolve identically for every agent, every department, every time. Without a semantic layer, a thousand agents produce a thousand truths, and the organization inherits the reconciliation cost.

Must 3 — Governed. Governance is not a feature added after the demo; it is the foundation. Permissions must be resolved before generation, approvals routed before execution, and policies enforced in the path of the query — not reported after the damage. A validation stage the agent can skip is not governance; it is decoration.

Must 4 — Auditable. Every generated query, every dry-run, every approval, every rollback must be on record, attributable, and replayable. "The AI did it" is not an acceptable line in an audit report. Work must be shown — always.

Must 5 — Cost-controlled. Every query must be estimated before execution: bytes scanned, monetary cost, runtime. Budgets must be enforced with circuit-breakers, not discovered on invoices. When machines can spend money at machine speed, cost control at human speed is no control at all. The industry's 2026 pivot toward cost- and energy-per-workflow as the primary optimization target has moved this principle from the end of the list to the sharpest edge of the standard; a proposal to extend it explicitly to energy awareness (clause

C5.5) is in public review.

3. What an ODAP Is — and Is Not

An **Open Data Agent Platform** is the single, governed entry point where AI agents interact with enterprise data, enforcing all Five Musts in the path of every query.

The boundary matters as much as the definition. An ODAP is **not a Text-to-SQL tool**: generation is the easy twenty percent, and it commoditizes further with every model release; an ODAP assumes generation and governs it. It is **not a BI dashboard**: dashboards show humans what happened, while an ODAP decides what machines are allowed to do. It is **not a data catalog**: catalogs describe data, while an ODAP stands in the path of access and enforces the rules. And it is **not an observability tool**: observability tells you on Tuesday what Monday's query cost; an ODAP tells the query "no" before it runs.

The composite failure modes are equally instructive. Open access without governance is a breach waiting to happen. Governance without openness is vendor lock-in wearing a badge. Both together without cost control is a blank check signed by a machine.

4. Reference Architecture

The Five Musts are principles; the five-layer reference architecture describes how a platform implements them.

Layer 1 — Universal Data Gateway. Connectivity to warehouses, lakes, object storage, and open table formats, with documented exit paths. (Implements: Open.)

Layer 2 — Smart Semantic Layer. Business metrics, dimensions, entities, and policies defined once and resolved consistently, with ownership and versioning. (Implements: Semantic.)

Layer 3 — Multi-Agent SQL Engine. A staged pipeline — planning, generation, validation, execution, repair — in which no stage can be skipped by the agent. Validation encompasses syntax, permissions, security, and cost estimation. (Implements: Governed, and the enforcement half of Cost-controlled.)

Layer 4 — Persistent Memory. Learning from corrections, query history, and organizational context across sessions. Memory is an architecture enhancement that improves accuracy over time; it is not an entry criterion for category membership.

Layer 5 — Governance & Observability. Audit logs, lineage, approval records, budget circuit-breakers, sandbox validation, quality scoring. (Implements: Auditable and Cost-controlled, and provides the evidentiary substrate for Governed.)

User or Agent
→ natural-language request
→ semantic context + policy + memory
→ planner → generator → validator → executor → reviewer
→ open data / warehouses / lakehouse tables / APIs
→ answer + table + chart + confidence + audit trail

5. Measuring Conformance: The Scoring Rubric

Principles that cannot be scored cannot be adopted, challenged, or improved. The ODAP Scoring Rubric v1.0 decomposes each Must into three to four assessable clauses — twenty in total, identified O1.1 through C5.4 — each anchored on a three-point scale: 1 (capability absent or purely manual), 3 (present but optional, incomplete, or not enforced by default), and 5 (enforced by default, in the path of every query). A dimension's score is the rounded mean of its clauses; a platform qualifies as an ODAP when every dimension scores at least 3; the maximum total is 25.

Two properties distinguish the rubric from vendor marketing checklists. First, its anchors reward default enforcement: a capability that exists but can be bypassed scores 3, not 5, because a gate with a side door is not a gate. Second, it is publicly challengeable: any score or clause may be contested by filing an issue citing the clause identifier and public evidence, with a committed fourteen-day response window and credited corrections. The rubric's full text is published under CC0 and versioned; clauses may be refined in minor versions, while the Five Musts themselves change only by major version with community consensus.

6. The Landscape: Applying the Rubric in Public

In mid-2026 the rubric was applied to five vendor archetypes, producing the ODAP Landscape 2026. Databricks scored highest (23/25), strong across governance, auditability, and — following its DAIS 2026 releases — cost control; a public recalibration proposing 24/25 is in its challenge window at the time of writing. Google's Agentic Data Cloud scored 22/25, formidable on governance and cost enforcement, with openness bounded by its own ecosystem. Starburst (18/25) leads on open-data federation while trailing on pre-execution cost guardrails. Acceldata (16/25) is a strong governance and observability control plane that is not yet a query gateway. Generic agent-first cloud platforms (12/25) excel at orchestration but lack semantic grounding and execution governance — speed without brakes.

The headline finding is that **no platform scores 25**. Every vendor is strong on some Musts and silent on others; the most consistent industry-wide gap is the semantic dimension, where business-editable metric definitions with ownership and versioning remain unshipped by every major platform assessed. The category is being assembled in real time, from different directions, by vendors with different incentives — which is precisely why its definition should not belong to any one of them.

The Landscape is republished as the market moves. Its first recalibration was triggered within days of a major vendor keynote, with every proposed score change cited to a clause and opened to challenge — a demonstration that the standard responds to events rather than aging as a PDF.

7. Reference Implementation

A standard without a running implementation is a lecture. The ODAP reference implementation, **odap-gateway** (Apache-2.0), is an open-source governed gateway for AI agents querying enterprise data, deliberately scoped to a single source done deeply — Google BigQuery — before breadth: schema-aware SQL generation grounded in approved tables and fields, mandatory dry-run cost estimation before every execution, threshold-based approval routing, and a full audit trail over query runs and governance decisions, built on a

staged pipeline in which generation, validation, and review are separate, non-skippable stages.

Its defining artifact is the clause-by-clause conformance mapping published alongside the code: at v0.1.0, four clauses are Implemented, twelve Partial, and five Planned — statuses that were deliberately downgraded during verification wherever behavior fell short of the anchor. The mapping names its gaps: no sandbox execution yet, no organization-level hard circuit-breaker yet, no business-editable semantic layer yet. This honesty is not modesty; it is method. A reference implementation that overstates conformance corrupts the rubric it exists to demonstrate. Gaps are roadmap, and the roadmap is public.

8. Governing the Standard Itself

A standard's credibility is its process. The ODAP texts — manifesto, rubric, landscape, and this paper — are maintained in public repositories under CC0, with changes proposed, challenged, and merged through recorded issues. Two such processes were live at the time of writing: a landscape recalibration following a vendor's product announcements, and a rubric extension proposing energy awareness (C5.5) as an emerging clause, triggered by peer-reviewed research and scored-but-not-averaged so that vendors are not penalized for a capability the market is only beginning to ship.

The standard has its first institutional adopter: the CloudSystem Association formally adopted the Five Musts and Rubric v1.0 as the evaluation framework for its Intelligent SQL Workflows reference-architecture program (Resolution CloudSystem-TC-2026-001), with the resolution and a conflict-of-interest disclosure published together in its public governance archive. The disclosure is worth underlining as practice: the ODAP initiative and the adopting association share a founder, and the adoption resolution states explicitly that the decision rests on the standard's openness, clause-level rigor, and challenge mechanism — with the affiliation published rather than hidden. Scrutiny is part of the standard too.

9. Future Work

Three directions are open. **The semantic layer** is the industry's most consequential unfilled slot: no assessed platform ships business-editable metric definitions with ownership and versioning, and the standard's S2 clauses define the target for whoever closes it. **Resource and energy governance** extends the fifth Must from dollars to joules as clause C5.5 matures from emerging to averaged. **Agent portability** — whether agent definitions, workflows, and memory can move between platforms in open formats — is a natural extension of the openness dimension as protocol-level standardization (MCP, A2A) advances, and is under consideration for a future working-group position paper.

10. An Invitation

This standard is a starting point, not a finish line. Data engineers, platform architects, CFOs who have opened a surprise invoice, CISOs who have traced an incident to an unlogged query, vendors, and standards bodies are invited to challenge, extend, and adopt the Five Musts. The texts are CC0; the challenge process is public; corrections are credited.

Argue with us. Improve this. Build to it.

Every query a machine writes should pass through a gate a human can trust.

Appendix A — Rubric Clause Index

Open: O1.1 open table formats • O1.2 heterogeneous sources • O1.3 exit cost • O1.4 protocol openness.

Semantic: S2.1 metric definition • S2.2 consistent resolution • S2.3 context awareness • S2.4 nested model support. **Governed:** G3.1 pre-generation permissions • G3.2 approval routing • G3.3 pipeline separation •

G3.4 sandbox validation. **Auditable:** A4.1 generation trail • A4.2 decision trail • A4.3 explainability • A4.4

replayability. **Cost-controlled:** C5.1 pre-execution estimation • C5.2 budget enforcement • C5.3 attribution granularity • C5.4 optimization loop • C5.5 resource & energy awareness (v1.1, proposed).

Appendix B — Canonical Sources

Manifesto: googlesql.com/odap/manifesto • Standard & rubric:

github.com/open-data-agent-platform/awesome-odap • Reference implementation:

github.com/open-data-agent-platform/odap-gateway • Institutional adoption archive:

cloudsystem.org/standards/governance • Landscape 2026 reports (EN/ZH): [awesome-odap /reports](#).

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